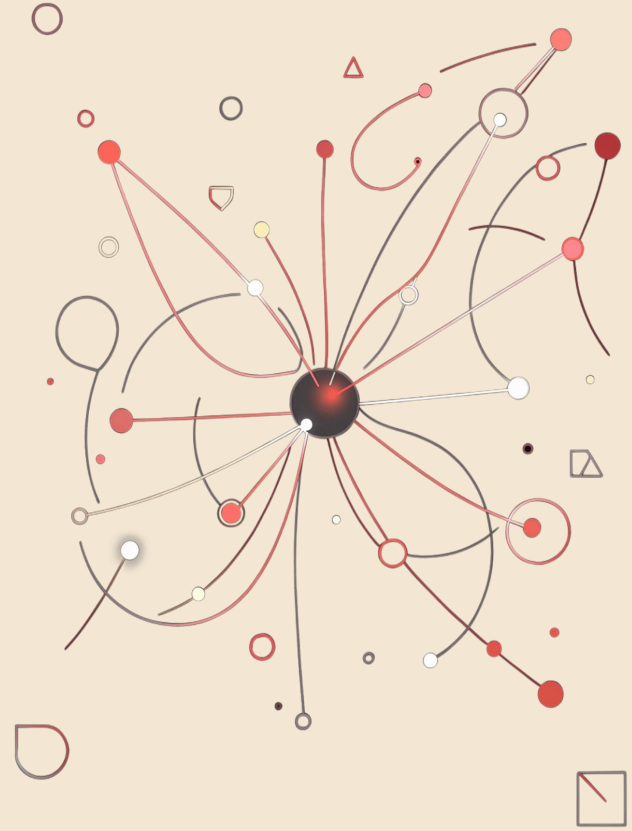


K-Nearest Neighbors (KNN) in Machine Learning

A Supervised Learning Algorithm for Classification and Regression





What is KNN?

Supervised Learning

Learns from labeled training data to predict outcomes

Lazy Learner

No explicit training phase — stores all data and computes at prediction time

Instance-Based

Predictions made by comparing new data to stored examples

"K" Neighbors

K = number of nearest neighbors used to vote on the prediction

Real-World Examples

🎬 Movie Recommendations | 🏥 Medical Diagnosis | ✍️ Handwriting Recognition | 👤 Face Recognition | 🛒 Product Recommendations

How KNN Works — Step by Step



Store Training

Receive Query

Compute Distances

Select K Neighbors

KNN makes no assumptions about data distribution — it simply measures proximity at prediction time, finds the K closest training examples, and returns the majority class (classification) or mean value (regression).

Distance Metrics

Euclidean Distance

Straight-line distance — most commonly used for continuous data.

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Manhattan Distance

Sum of absolute differences — ideal for grid-based or high-dimensional data.

$$d = \sum_{i=1}^n |x_i - y_i|$$

Minkowski Distance

Generalized metric — unifies Euclidean (p=2) and Manhattan (p=1).

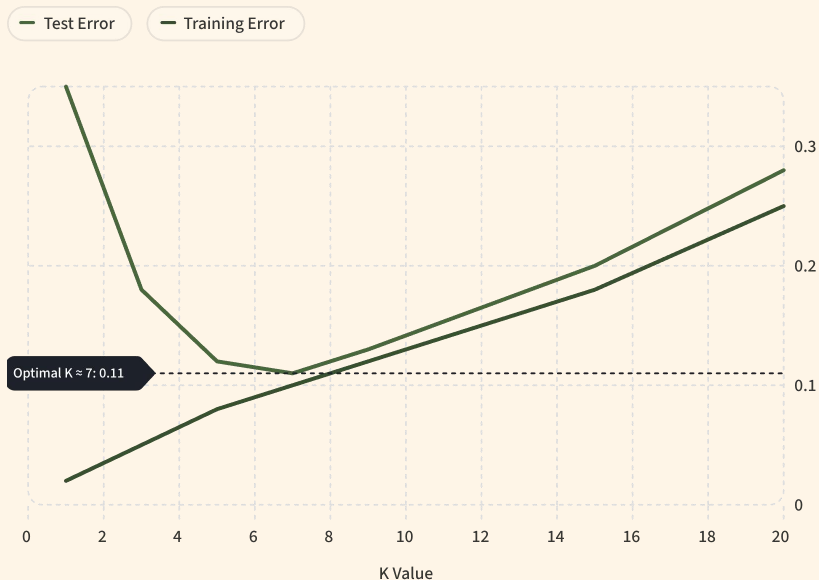
$$d = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

Hamming Distance

Counts differing positions — used for categorical or binary features.

$$d = \frac{1}{n} \sum_{i=1}^n (x_i \neq y_i)$$

Choosing the Optimal K



Small K (e.g. K=1)

Sensitive to noise, high variance → risk of **overfitting**

Large K (e.g. K=20)

Smooth boundary, high bias → risk of **underfitting**

Sweet Spot

Use **cross-validation** or **grid search** to find the optimal K at the error valley

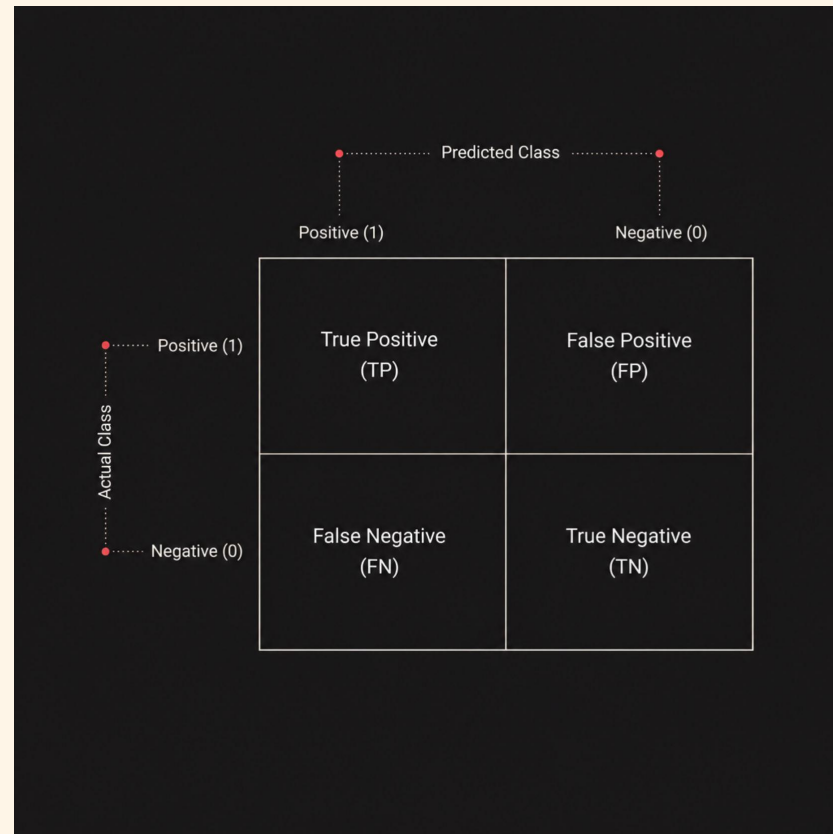
Model Evaluation Metrics

Classification Metrics

- **Accuracy, Precision, Recall, F1 Score** — measure correctness across classes
- **Confusion Matrix** — visualizes TP, TN, FP, FN counts
- **ROC Curve & AUC** — measures discriminative power across thresholds

Regression Metrics

- **MAE & RMSE** — average prediction error magnitude
- **R² Score** — proportion of variance explained by the model



Advantages & Limitations

✓ Advantages

- Simple, intuitive, and requires **no training phase**
- Works for both **classification and regression**
- Handles **multi-class problems** naturally

⚠ Limitations & Challenges

- Slow prediction on large datasets; **high memory usage**
- Sensitive to noisy data and **irrelevant features**
- Requires **feature scaling** (e.g. normalization)
- **Curse of Dimensionality:**
Performance degrades as feature count grows



Real-World Applications



Healthcare

Disease diagnosis and medical image classification



Finance

Credit risk prediction and fraud detection



Retail

Product recommendations and customer segmentation



Computer Vision

Face recognition and image classification



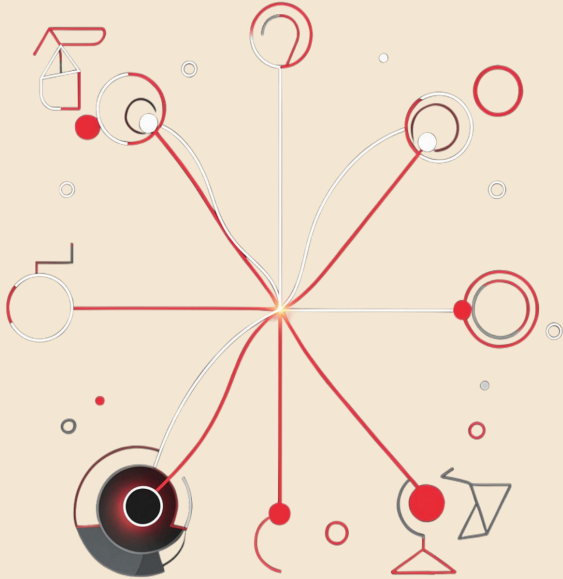
Cybersecurity

Intrusion detection and spam classification

KNN vs. Logistic Regression

Criterion	KNN	Logistic Regression
Learning Type	Instance-Based (Lazy)	Model-Based (Eager)
Training Time	None ($O(1)$)	Moderate
Prediction Time	Slow ($O(n \cdot d)$)	Fast ($O(d)$)
Interpretability	Low	High
Scalability	Poor on large data	Good
Decision Boundary	Non-linear, flexible	Linear
Best Use Case	Small datasets, multi-class	Binary classification, large data

Summary & Key Takeaways



- KNN is a **supervised, lazy, instance-based** learning algorithm — no model is built during training
- Uses **distance metrics** to find the K nearest neighbors and predict via majority vote or mean
- Choice of **K** and **distance metric** critically affects bias-variance tradeoff and model performance
- Widely applied in **healthcare, finance, retail, computer vision, and cybersecurity**

"KNN predicts the future by learning from the closest examples of the past — one of the simplest yet most effective ML algorithms."